



1.13

Function Model Selection and Assumption Articulation

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 1 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Student Information

Name: _____ Class: _____ Date: _____

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: Model choice should combine context, rate patterns, residuals, and domain.

2. Write one precise sentence explaining this principle: Residual = observed - predicted. Random scatter around zero supports plausibility.

3. Write one precise sentence explaining this principle: A strong fit does not prove causation or guarantee valid extrapolation.

4. Write one precise sentence explaining this principle: Assumptions should address constancy of parameters and meaningful input range.

Core

5. A table has nearly constant ratios for equal input steps. Which model is most plausible?

6. A residual plot forms a clear U-shape. What does that say about a linear model?

7. Give one assumption behind an exponential population model.

8. State what the formula $e_i = y_i - \hat{y}_i$ tells you and name any required restriction.

9. State what the formula absolute error = $|e_i|$ tells you and name any required restriction.

10. State what the formula relative error = $\frac{|e_i|}{|y_i|}$ tells you and name any required restriction.

Medium

11. Select or assess a model for the evidence: 2,5,8,11,14 at equal steps.

12. Select or assess a model for the evidence: 3,6,12,24,48 at equal steps.

13. Select or assess a model for the evidence: 1,4,9,16,25 at consecutive integers.

14. Select or assess a model for the evidence: residuals 1.2, -0.9, 0.7, -1.1, 0.8.

15. Select or assess a model for the evidence: residuals $-4, -1, 2, 4, 2, -1, -4$.

Hard

16. Observed values are 20,31,48,70; model A predicts 21,30,49,69 and model B predicts 18,34,46,72. Compare MAE.

17. A quadratic and exponential model both fit four data points closely. Name two additional criteria for choosing.

18. Why is extrapolating a high-degree polynomial far beyond data risky?

19. Give a situation where underprediction is more serious than equal-sized overprediction.

Difficult

20. Model A residuals have mean 0 and strong alternating pattern. Model B residuals have mean 0.4 and random scatter. Which is more plausible?

21. A model has $R^2 = 0.999$. Give three reasons it may still be inappropriate.

22. Compute relative errors for observed 4 and 400 when both predictions are 2 units low. Compare.

Challenge / HOT

23. Construct a small dataset for which both a linear and quadratic model interpolate all points.

24. Explain why residual plots are representation transformations rather than new data.

AP-Style Multiple Choice

25. Random residual scatter around zero generally
(A) supports plausibility

- (B) proves causation
 - (C) guarantees extrapolation
 - (D) requires a quadratic
26. Residual is usually
- (A) predicted-observed
 - (B) observed-predicted
 - (C) observed/predicted
 - (D) squared error
27. Constant ratios suggest
- (A) linear
 - (B) quadratic
 - (C) exponential
 - (D) logarithmic only
28. A U-shaped residual pattern for a linear model indicates
- (A) good fit
 - (B) missed curvature
 - (C) perfect prediction
 - (D) causation
29. Which is an assumption, not evidence?
- (A) residuals are random
 - (B) growth rate remains constant in future
 - (C) MAE is 2
 - (D) data are increasing
30. Overfitting means
- (A) too few parameters
 - (B) model follows noise rather than general structure
 - (C) all residuals equal
 - (D) the model is linear

AP-Style Free Response

- 31.** A dataset is $(0, 5), (1, 8), (2, 13), (3, 20), (4, 29)$. (a) Compare first and second differences. (b) Select a model family. (c) Construct the model. (d) State one assumption and one limitation.

- 32.** Two models predict the same observed data. Model L residuals are $-2, -1, 0, 1, 2$; Model E resid-

uals are $1, -1, 0, -1, 1$. (a) Compute MAE. (b) Discuss pattern. (c) Recommend a model.

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.